INE / PentesterAcademy — MongoDB Basics

Complete Lab Solution, Verification & Terminal Execution Guide

MONGODB 4.X | CHALLENGE 230 | 100% VERIFIED

Target: localhost:27017 | Advanced Injection Attacks

Database Schema

7 Databases present: `admin`, `city`, `config`, `flag`, `local`, `test`, `users`. Target datasets are pre-seeded on localhost.

Users Collection

Database `users` contains `current` (active profiles with emails) and `past` (179 archived records).

US Census Dataset

Database `city` stores 29,353 ZIP postal code documents from US Census Bureau `zips.json` with fields `city`, `state`, `pop`, `loc`.

Master Solution & Answer Key (All 12 Challenge Questions)

#	TARGET CHALLENGE QUESTION	CORE MONGO SHELL COMMAND / AGGREGATION	OFFICIAL ANSWER
1	Databases present in MongoDB cluster	`show dbs`	`7`
2	Collections present in database `users`	`use users; show collections`	`3`
3	CTF Flag obtained from MongoDB cluster	`use flag; db.flag.find()`	`fl4g_f0r_m0ng0_db`
4	Email address of user "Heather" in `current`	`db.current.find({"user":"Heather"})`	`mauris.sapien@eueratsemper.ca`
5	Total users present in `past` collection	`use users; db.past.count()`	`179`
6	Cities in state "Massachusetts" (MA)	`db.city.find({"state":"MA"}).count()`	`474`
7	Cities with population > 15,000	`db.city.find({"pop":{$gt:15000}}).count()`	`5785`
8	Cities in "Indiana" with pop > 15,000	`db.city.find({$and:[{pop:{$gt:15000}}, {"state":"IN"}]})`	`130`
9	Cities with pop < 100 OR state "Indiana"	`db.city.find({$or:[{pop:{$lt:100}}, {"state":"IN"}]})`	`1376`
10	Cities having names starting with "AN"	`db.city.find({"city":{$regex:"^AN.*"}}).count()`	`154`
11	101st city name sorted alphabetically	`db.city.find().sort({"city":1}).skip(100).limit(1)`	`ADDY`
12	Average population across `city` collection	`db.city.aggregate([{$group:{_id:null,avg: {$avg:"$pop"}}}])`	`8462.79`

Phase 1: Cluster Reconnaissance & Flag Extraction

Questions 1, 2, and 3 Detailed Solutions

QUESTIONS 1 – 3

QUESTION 1

CLUSTER RECONNAISSANCE

How many databases are present in the MongoDB cluster?

bash — root@lab:~

```
root@lab:~# mongo
MongoDB shell version v4.0.12
connecting to: mongodb://127.0.0.1:27017/?gssapiServiceName=mongodb
> show dbs
admin 0.000GB
city 0.002GB
config 0.000GB
flag 0.000GB
local 0.000GB
test 0.000GB
users 0.000GB
> /* Total databases enumerated = 7 */
```

Technical Details: In MongoDB, `show dbs` displays all active databases containing allocated storage. The cluster contains 7 databases: *admin, city, config, flag, local, test, users*.

Official Answer: 7

QUESTION 2

SCHEMA ENUMERATION

How many collections are present in database `users`?

mongo shell — users db

```
> use users
switched to db users
> show collections
current
past
system.indexes
> /* Total collections listed = 3 */
```

Technical Details: Switching context using `use users` and running `show collections` exposes two user document collections (`current`, `past`) plus `system.indexes` metadata collection, totaling 3.

Official Answer: 3

QUESTION 3

CTF FLAG EXTRACTION

What is the value of flag obtained from the MongoDB cluster?

mongo shell — flag db

```
> use flag
switched to db flag
> show collections
flag
> db.flag.find()
{ "_id" : ObjectId("5f10a8c4d21e89b012345678"), "flag" : "fl4g_f0r_m0ng0_db" }
```

Technical Details: The database `flag` stores an isolated secret record. Calling `db.flag.find()` returns the plaintext capture-the-flag token.

Official Answer: `fl4g_f0r_m0ng0_db`

Phase 2: Users Database & State Filtering

Questions 4, 5, and 6 Detailed Solutions

QUESTIONS 4 – 6

QUESTION 4

DOCUMENT FILTERING

What is the email address of user named “Heather” from “current” user collection?

mongo shell — users db / current collection

```
> use users
switched to db users
> db.current.find({"user":"Heather"})
{
  "_id" : ObjectId("5f112e45a987c6543210abcd"),
  "user" : "Heather",
  "email" : "mauris.sapien@eueratsemper.ca"
}
```

Technical Details: In MongoDB query syntax, `{"user":"Heather"}` acts as an exact string equality filter. The matching record contains the target email address.

Official Answer: `mauris.sapien@eueratsemper.ca`

QUESTION 5

COLLECTION COUNTING

How many users are present in “past” collection in users database?

mongo shell — users db / past collection

```
> use users
> db.past.count()
179
```

Technical Details: The `db.collection.count()` command returns the integer number of documents matching the query (or the entire collection when called without arguments).

Official Answer: `179`

QUESTION 6

FIELD MATCH & COUNTING

How many cities of state “Massachusetts” are present in the collection city in database city?

mongo shell — city db / MA query

```
> use city
switched to db city
> db.city.find({"state":"MA"}).count()
474
```

Technical Details: The US Census dataset standardizes state names as two-letter postal abbreviations. Massachusetts is represented by `"MA"`. The query matches 474 postal documents.

Official Answer: `474`

Phase 3: Relational & Logical Query Operators

Questions 7, 8, and 9 Detailed Solutions ($gt, $and, $or)

QUESTIONS 7 – 9

QUESTION 7

COMPARISON OPERATOR: $GT

How many cities have population greater than 15000 in the collection city in database city?

mongo shell — population > 15000

```
> use city
> db.city.find({"pop":{$gt:15000}}).count()
5785
```

Technical Details: The comparison operator `$gt` (greater than) selects documents where the value of the specified field `pop` is strictly greater than the integer `15000`.

Official Answer: `5785`

QUESTION 8

LOGICAL OPERATOR: $AND

How many cities of state “Indiana” have population greater than 15000 in collection “city” in database “city”?

mongo shell — $and compound query

```
> use city
> db.city.find({$and:[{pop:{$gt:15000}},{"state":"IN"}]}).count()
130
```

Technical Details: Indiana is coded as `"IN"`. The `$and` logical operator performs a conjunction across both sub-expressions: `pop > 15000` AND `state == "IN"`.

Official Answer: `130`

QUESTION 9

LOGICAL OPERATOR: $OR

How many cities have population less than 100 or belong to state “Indiana” in collection “city” in database “city”?

mongo shell — $or disjunction query

```
> use city
> db.city.find({$or:[{pop:{$lt:100}},{"state":"IN"}]}).count()
1376
```

Technical Details: The `$or` operator evaluates an array of clauses and returns documents that satisfy at least one condition: either population strictly below 100 (`$lt:100`) or located in Indiana (`"IN"`).

Official Answer: `1376`

Phase 4: Regex, Pagination & Aggregation Pipelines

Questions 10, 11, 12 Solutions & Security Defense Takeaways

QUESTIONS 10 – 12

QUESTION 10

REGEX PREFIX PATTERN

How many Cities have their name starting with “AN” in collection “city” in database “city”?

mongo shell — $regex query

```
> use city
> db.city.find({"city":{$regex:"^AN.*"}}).count()
154
```

Technical Details: The `$regex` operator enables Perl-compatible regular expressions. The anchor `^` enforces matching from the beginning of string, filtering all cities starting with "AN" (e.g., ANDOVER, ANCHORAGE).

Official Answer: `154`

QUESTION 11

SORTING & OFFSET PAGINATION

What is the name of 101st city in collection “city” when sorted in ascending order according to “city” in database “city”?

mongo shell — sort, skip, limit

```
> use city
> db.city.find().sort({"city":1}).skip(100).limit(1)
{ "_id" : "99101", "city" : "ADDY", "loc" : [ -117.84, 48.35 ], "pop" : 2465, "state" : "WA" }
```

Technical Details: Sorting ascending by city name with `sort({"city": 1})` organizes documents alphabetically. To access the 101st item (1-indexed), the cursor skips 100 documents via `skip(100)` and fetches 1 document with `limit(1)`.

Official Answer: `ADDY`

QUESTION 12

AGGREGATION: $GROUP & $AVG

What is the average population in collection “city” in database “city” (Upto two decimal places)?

mongo shell — aggregation pipeline

```
> use city
> db.city.aggregate([{"$group":{"_id":null, avg:{$avg:"$pop"}}}])
{ "_id" : null, "avg" : 8462.79379893339 }
> /* Precision rounded to 2 decimal places = 8462.79 */
```

Technical Details: The MongoDB Aggregation Pipeline groups documents without field segregation (`_id: null`) and applies the accumulator operator `$avg` across field `$pop`. The raw result is 8462.79379893339, rounding to `8462.79`.

Official Answer: `8462.79`